

Haoyu (Howie) HAN

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PROFESSIONAL SUMMARY

Master's graduate with hands-on experience in structural analysis, CAD modelling, and rapid prototyping with additive manufacturing. Strong background in robotics and control engineering, enabling performance-driven design and system-level thinking. Experienced in full product development cycle and participation in various engineering projects, contributing to design optimisation, integration, and technical decision-making. Passionate about technological innovation and adept at collaborating across disciplines to deliver robust, manufacturable, and commercial engineering solutions.

CORE SKILLS

- **Mechanical Engineering:** SolidWorks, OnShape, AutoCAD (GB&T, BOM), Fusion 360; Reverse Engineering (3D Scanning, QuickSurface); CNC Machining, Additive Manufacturing / 3D Printing; DfM/DfAM.
- **Mathematical Modelling & Simulation:** MATLAB, SolidWorks Simulation, Adams, COMSOL Multiphysics.
- **Mechatronics & Embedded Systems:** C/C++, Python, Linux; Arduino, ESP32, STM32.
- **Languages:** English (IELTS 7.0), Mandarin (Native), German (A1).

EDUCATION

National University of Singapore MSc. in Robotics Singapore	08/2025 – 01/2027
Beihang University (BUAA) B. Eng in Robotics Beijing, China	09/2021 – 06/2025

WORK EXPERIENCE

Robotics and Additive Manufacturing R&D Intern Alstom (Singapore) Innovation Station	2026.01 - 2026.07 Singapore
• Inspection Robot Development: Spearheaded the end-to-end mechanical design of a metro train inspection robot using <i>OnShape</i>. Applied DfAM principles with PA6 engineering plastics to create a support-less, integrated structure, successfully consolidating cable clips, control panel damper, and quick-release mechanisms into just over 10 parts. • Product Delivery: Delivered a fully functional product weighing 5 kg, capable of traversing 85° inclines and 15 cm obstacles with 2-3 hours of continuous operation, achieving zero external supplier dependency for structural components. • Reverse Engineering & Supply Chain Optimisation: Utilised 3D scanning and <i>QuickSurface</i> to reverse-engineer obsolete transit spare parts. Successfully established an internal "R&D-to-Workshop" additive manufacturing pipeline, delivering 46 core components across 3 categories. This initiative eliminated traditional procurement lead times and reduced per-unit manufacturing costs by 90%. • Compliance & Workflow Automation: Authored a 10,000+ word SOP for long-term 3D printer maintenance, aligning strictly with EHS requirements. Developed an automated project management dashboard via Power Apps, improving cross-departmental communication efficiency by nearly 30%.	
Medical Mechatronics Research Intern Duke-NUS Medical School	2025.08 - 2026.01 Singapore
• Mechanism Simplification: Translated complex human ribcage structure and movements into a realisable engineering prototype. Conducted kinematic and dynamic analysis to design a compact rack-and-gear drive for motion simulation. • DfAM Prototyping: Applied DfAM principles to integrate PLA-TPU multi-material joints, optimising parameters (interlocking, wall thickness, infill, etc.) to drastically reduce structural components from 62 pcs to 5 pcs and successfully decreased the print failure rate by 20%. • Low-Level Servo Control Integration: Directed the selection of mechatronic components and constructed a low-level servo control system utilising Arduino and Dynamixel XM430 industrial servos. Collaborated to develop serial communication protocols, achieving full closed-loop motion control that satisfied physiological tracking metrics.	
Robot Testing Intern AUBO (Jiangsu) Robotics Co., Ltd.	2024.07 - 2024.09 Jiangsu, China
• System Testing: Conducted software testing for 6 motion-planning components of a 5 kg 6-axis robotic arm in Linux, assessing 8 performance metrics (repeat positioning accuracy, collision detection, I/O interfaces, etc.); reported 12 critical bugs and cooperated with the R&D department to fix them all within one month.	

PORJECT EXPERIENCE

Magnetically-Driven Left Ventricular Assist Device (LVAD) Design & Simulation 2024.09 - 2025.06

Undergraduate Thesis | Supervisor: Assoc. Prof. Cai Yueri, Beihang University Beijing, China
(Design schemes and simulation outcomes contributed to the feasibility validation of a PhD research programme)

- **Device Design:** Addressed clinical requirements for ventricular assist applications by constructing a biomimetic flexible LVAD pump chamber prototype, integrating a unidirectional valve assembly within *SolidWorks*.
- **Multiphysics Modelling & Simulation:** Developed a comprehensive magneto–solid–fluid coupled framework in *COMSOL Multiphysics*. Applied a Neo-Hookean hyperelastic constitutive model for magneto-responsive material deformation, calculated magnetic driving forces via Maxwell stress tensors, and integrated Arbitrary Lagrangian-Eulerian (ALE) methods for dynamic Fluid-Structure Interaction (FSI) analysis.
- **Clinical Metric Validation:** Verified myocardium-like stable contraction and unidirectional flow under alternating magnetic fields through transient simulations. Achieved a stroke volume of 62.5 mL, a cardiac output of 3.75 L/min, and an ejection fraction exceeding 65%, satisfying ISO standards for medical devices and verifying preclinical feasibility.

High-Precision Omnidirectional Steering-Wheel Chassis R&D 2023.09 - 2024.05

Core Mechanical Designer | Team won National 1st Prize in the 23rd ROBOCON Beijing, China

- **Mechanical Design:** Conducted overall scheme definition and mechanical design of a dual-motor driven steering wheel module in *SolidWorks*, featuring axial bevel gear transmission and steering spur gear reduction mechanisms. Practised static and dynamic simulations using *SolidWorks Simulation* and *Adams* to validate gear meshing and core load-bearing components, assisting in precise actuator, gear, and bearing selection.
- **Engineering Drawing & Tolerancing:** Drafted manufacturing-ready 2D engineering drawings using *AutoCAD* for CNC outsourcing, enforcing strict GB tolerance standards (e.g. 7 gear accuracy qualification). Coordinated advanced manufacturing techniques including FDM 3D printing and carbon fibre sheet cutting and independently assemble and maintain 10 production-ready omnidirectional wheels.
- **Chassis Integration & Joint Testing:** Optimised the overall layout and structural architecture of 2 mobile robot chassis, providing hardware routing support. Participated in an intensive 2-month joint mechatronic testing phase, delivering highly integrated steering modules (1.5 kg unit weight, 6 kg payload capacity) that achieved a steering resolution of 1.7°

Rigid-Flexible Coupling Tendon-Driven Robotic Spatial Wrist R&D 2022.09 - 2023.06

Core Mechanical Designer | Results adopted by subsequent Laboratory Research Beijing, China

- **Kinematic Analysis: & Modelling** Designed a 3 DoF serial wrist mechanism functioning as a spherical joint for an elderly-assisting mobile robot platform. Formulated the geometric kinematic model based on Denavit-Hartenberg (D-H) parameters; utilised *MATLAB Robotics Toolbox* and *SolidWorks Simulation* to analyse and optimise a hemispherical workspace (360° Yaw/Roll, 95° Pitch, and a 200 mm spatial span).
- **Tendon-Driven Mechanism & Adaptive Gripper:** Developed a novel rigid-flexible coupling adaptive gripper actuated by a bus servo coupled with a wire spool mechanism. Engineered an organic grabbing topology where line tension triggers the elastic deformation of flexible aluminium sheet metals, enabling stable and form-adaptive manipulation of dynamic, irregular everyday objects (e.g., eggs).
- **Structural Analytics & Physical Assembly:** Executed comprehensive FEA structural analysis in *SolidWorks Simulation* to calculate safety factor margins and fatigue lifespans of critical drive shafts and structural links. Managed the full prototyping loop via SLA/FDM 3D printing and precision CNC machining, assembling three 35 kg·cm dual-axis servos to guarantee a robust 1 kg load capacity with 0.5° angular resolution.

CO-CURRICULAR EXPERIENCE

Counsellor, International School, Beihang University 2024.07 - 2025.08

- Established Beihang International Students' Robotics Team and led team to 4 national championships within one year.
- Organized students to participate in over 8 innovation competitions and 2 gala events (one at United Nation level).
- Organized and delivered over 5 school-level science and technology lectures for international students.

President, Robot Association, Beihang University 2023.09 - 2024.06

- Organised large-scale robotics events (e.g., a 400+ participant robot competition) and managed collaborations with other universities, boosting community engagement and organizational influence.
- Led technical initiatives (e.g., a robotic dog prototype) and improved team operations through 12 workshops.